

Alpha↔Omega Dynamics
The Hidden Temporal Dynamics of Stokes
Manual 43°C

Reginald Poyau

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1 Introduction

1.1 Purpose

Manual 43°C records worked AlphaOmega Dynamics calculations, reported comparisons, tables, and figure witnesses. The derivational authority is the main note [?].

Each chapter fixes a cited row, carries the AOD calculation, and reports the resulting comparison against cited data where used.

2 Core worked rows

2.1 Loss-channel worked table

A shell-qualified motif window is fixed first, then read through the loss-channel, dominant residual-duration class, support mass, and attenuation machinery of the main note.

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Table 1: Loss-channel worked rows on shell-qualified motif windows.

Row	T_{eval} [bip]	$N_m = (n_1, n_2, n_3)$	B^*	Λ_b	R_{loss} [trit/bip]	$R_{\text{loss}}^{(\text{biz})}$ [biz]
L1	12	(2, 4, 6)	5	4	$(2 + 4\ell_2)/12$	$(3^{(2+4\ell_2)/12} - 1)5^{-4}$
L2	24	(1, 8, 15)	137	6	$(1 + 8\ell_2)/24$	$(3^{(1+8\ell_2)/24} - 1)137^{-6}$

$\ell_2 := \log_3(3/2)$. The cited ledger resolves admissibility counts, dominant residual-duration class, support mass, attenuation, and the loss-channel witness on one row.

Use the loss-channel and attenuation machinery of the main note on the same cited ledger.

2.2 B^* ladder worked table

The dominant residual-duration class is read after the cited sample ladders are fixed; the Jensen-gap witness is preserved by delaying averaging until after histogram formation.

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Table 2: B^* ladder worked rows.

Row	$b(\kappa)$ samples	H_B support	B^*
B1	(2, 2, 2, 4)	$H_B(2) = 3, H_B(4) = 1$	2
B2	(5, 5, 5, 7, 7)	$H_B(5) = 3, H_B(7) = 2$	5
B3	(137, 137, 139)	$H_B(137) = 2, H_B(139) = 1$	137

The three rows display a minimal two-cycle ladder, a four-cycle enclosure ladder, and a high-rung attractor witness. Histogram weighting and attenuation are formed on the cited sample set before any averaging.

2.3 Collision / export worked table

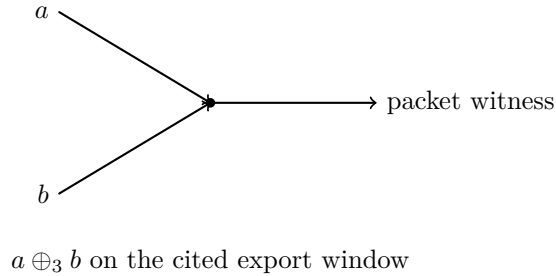
The cited export window is fixed first. Push/pull sign, effective-charge proxy, packet witness, and cut residual are then reported on that same ledger.

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Table 3: Collision / export worked rows.

Row	Inputs (a, b)	$a \oplus_3 b$	Push/pull	q_{eff}	Packet witness	$\mathcal{R}_{\text{cut}}$
X1	(+1, -1)	0	balanced	0	none	0
X2	(+1, +1)	-1	push	+1/2	emitted packet	0
X3	(-1, -1)	+1	pull	-1/2	emitted packet	0

Collision / export claims are valid only if the same window certifies the packet witness and the cut residual vanishes within tolerance.

**Figure 1:** Diagrammatic rendering of the cited collision / export row.

Use the collision / export, push/pull, effective-charge, and cut-residual machinery of the main note.

2.4 Field-property schema row

One shell-qualified AO field instance is fixed first and then reported through cadence spectrum, dominant residual-duration class, loss-channel witness, and torsional witness.

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Table 4: Field-property schema row.

Field property	Reported value
AO field key	O-tetrad:3:4:6
TTL	24 bip
Cadence spectrum	$\{1/4, 3/8, 5/12\}$ [cyc/bip]
B^* / Λ_b	137/6
R_{loss}	$(1 + 8\ell_2)/24$ [trit/bip]
$R_{\text{loss}}^{(\text{biz})}$	$(3^{(1+8\ell_2)/24} - 1)137^{-6}$ [biz]
q_{eff}	+1/2
T_{tors}	3

The schema row remains in exact/default-form gauge on the cited ledger.

2.5 Short-window confinement row

A cited short shell-qualified ledger is fixed first and then read through shell signature, dominant rung, proximity, confinement witness, and transmutation flag.

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Table 5: Short-window confinement rows.

Row	Key	Shell signature	B^*	Proximity / confinement	Transmutation
C1	tetron:3:3:6	$(\lambda=3/2, 137, \rho_2, V_4)$	137	close / confined	0
C2	tetron:3:3:6	$(\lambda=3/2, 139, \rho_3, V_4)$	139	failed / unconfined	1

The same cited short ledger resolves shell signature, dominant rung, proximity status, confinement witness, and transmutation flag.

2.6 Duon / duad family row

A cited two-seat family ledger is fixed first and then reported through family key, push/pull sign, effective-charge proxy, export witness, and cadence witness.

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Table 6: Duon / duad family rows.

Row	Key	Push/pull	q_{eff}	Packet witness	Cadence
F1	duon:2:1:7	push	+1/2	emitted packet	1/3
F2	duad:2:2:7	balanced	0	none	1/4

The same cited ledger supports the family key, push/pull sign, effective-charge proxy, packet witness, and cadence witness.

2.7 Open-seat wavelet row

A declared open-seat two-seat family ledger is fixed first and then carried into the loss-channel, delay, and reference-count witnesses.

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Table 7: Open-seat wavelet cited rows.

Row	q_{eff}	$\hat{q}_{\text{eff}}$	η_{out}	η_{mid}	η_{tot}	f_0	B^*	R_{loss} [trit/bip]	n_{ref}
O1	-1	-1	1063	265	664	139	(1, $\frac{190}{233}$)	$\frac{278+1328\ell_2}{2048}$	1024
O2	$-\frac{1}{2}$	$-\frac{817}{1657}$	1418	352	885	1024	(1, $\frac{256}{313}$)	$\frac{391+907\ell_2}{2048}$	873
O3	0	$\frac{41}{1543}$	896	761	1657	2048	(1, $\frac{284}{375}$)	$\frac{505+681\ell_2}{2048}$	1541
O4	$\frac{1}{2}$	$\frac{419}{819}$	158	457	133	205	(1, $\frac{279}{338}$)	$\frac{410+931\ell_2}{2048}$	331
O5	1	1	1053	95	1338	1024	(1, $\frac{203}{229}$)	$\frac{261+1338\ell_2}{2048}$	767
			1403	128	1787	2048			1024
									871

Optional converted witnesses are carried only in the temporal-conversion chapter. The cited open-seat wavelet row remains authoritative.

3 Temporal conversions and mass witnesses

This chapter records how cited rows are carried into converted witness form. Duration is reported in seconds, rates in hertz, and effective speed only through $\beta = v/c$ with $c = 1$.

3.1 Conversion rules

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Table 8: Conversion classes and required cited-row quantities.

Conversion class	Required cited-row quantities
Worked-row comparison	cited key, cited window, shell signature, B^* , R_{loss} , cadence, and any declared witness columns
Temporal-unit conversion	cited key, cited window, loss/report rates, shell/export predicates, and declared dictionary-map id
Diagrammatic rendering	cited key, shell signature, cadence spectrum, confinement/support witnesses, and loss/report rates

Converted witnesses are attached only after the cited row is fixed on the same ledger.

3.2 Converted rows

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Table 9: Converted-row starting columns used by the public rendering layer.

Converted row class	Required cited-row starting columns
Comparison row	key, cited window, λ -band, shell signature, B^* , R_{loss} , $R_{\text{loss}}^{(\text{biz})}$, cadence, torsional witness
Temporal-unit row	key, cited window, shell/export predicate, loss/report rates, declared dictionary-map id
Diagrammatic row	key, shell signature, cadence spectrum, confinement witness, support signature, and loss/report rates

The converted-row layer records only quantities already present on the cited ledger.

3.3 Diagrammatic temporal conversion rendering

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A cited collision / export row may be displayed diagrammatically after that ledger is fixed.

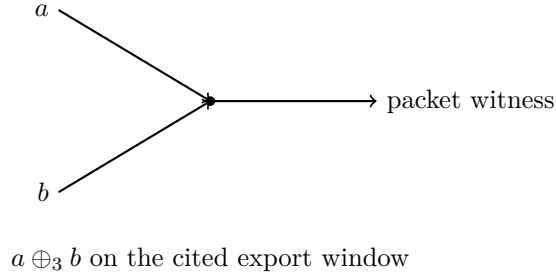


Figure 2: Diagrammatic rendering of the cited collision / export row.

The diagram is a later display of the same cited collision / export ledger.

3.4 Recurrent-ring rendering

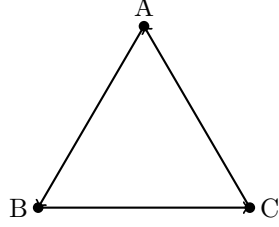
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A cited recurrent-ring row may be displayed as a projected three-node recurrent system.

Table 10: Recurrent-ring cited row.

Row	Topology	R_{loss} [trit/bip]	B^* [bip]	$\hat{\Lambda}$ [ticks]	$R_{\text{loss}}^{(\text{biz})}$ [biz]
E10.N1	3-node ring	0.4887777777777776	2	3	0.08885345920654222

CK mismatch and cut-residual diagnostics remain companion checks on the same cited recurrent-ring ledger.



projected recurrent ring on the cited ledger

Figure 3: Recurrent-ring rendering attached after the cited row is fixed.

The projected ring remains a display of the cited recurrent ledger rather than a replacement for it.

3.5 Decimal rendering

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The decimal table records convenience values only and does not replace the exact/default-form cited row.

Table 11: Decimal rendering values for the cited open-seat wavelet ledger.

Row	q_{eff}	$\hat{q}_{\text{eff}}$	η_{out}	η_{mid}	η_{tot}
E11.q-1	-1	-1.0	0.7496473906911142	0.7528409090909091	0.7502824858757062
E11.q-1/2	-1/2	-0.4930597465298733	0.4810267857142857	0.6254927726675427	0.5473747736873869
E11.q0	0	0.026571613739468567	0.2673684210526316	0.5187265917602997	0.44134802333117307
E11.q1/2	1/2	0.5115995115995116	0.5249169435215947	0.6217687074829932	0.5683760683760684
E11.q1	1	1.0	0.7505345687811832	0.7421875	0.748740906547286

Additional decimal companions such as R_{loss} , v , and n_{ref} remain in the companion source CSV and notebook.

3.6 Rest-mass witnesses and external comparisons

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The cited rows remain fixed first. The table below records downstream mass witnesses against cited comparison targets in the same reported unit family.

Table 12: Rest-mass witnesses and external comparisons.

Structural key	Cited-row quantities used	Witness quantity [MeV/ c^2]	External comparison target	Claim level
tetron:1:2:6	λ -band; B^* ; R_{loss} ; cadence	0.51099895069	electron (CODATA 2022)	mass comparison only
tetron:1:2:8	λ -band; B^* ; R_{loss} ; cadence	105.6583755	muon (CODATA 2022)	mass comparison only
tetron:1:2:10	λ -band; B^* ; R_{loss} ; cadence	1776.86	tau (CODATA 2022)	mass comparison only
tetron:3:3:6	shell signature; B^* ; confinement; R_{loss}	938.27208943	proton (CODATA 2022)	mass comparison only
tetrad:3:4:6	shell signature; B^* ; confinement; R_{loss}	939.56542194	neutron (CODATA 2022)	mass comparison only

Mass comparison only records a downstream numerical comparison. *Candidate identity* and *full identity claim* remain stronger downstream categories and are not asserted here.

These rows record numerical mass witnesses only; they do not by themselves assert a stronger identity claim.

4 Somatic mechanics across scales

This chapter records worked AOD witnesses across solar, galactic, atomic, biological, and recurrent ledgers.

4.1 Solar system

Anomalous perihelion precession of Mercury

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The cited 100-year solar-system ledger is fixed first and the perihelion witness is read as an extracted Mercury slice on that same long-window ledger.

Table 13: Solar perihelion cited row.

Row	Slice key	T_{eval} [bip]	B^*	Λ_b	R_{loss} [trit/bip]	T_{tors}
S1	tetron*: [sun,mercury] :6	36525	137	6	$\frac{1+8\ell_2}{24}$	3

The cited row fixes the shared long-window ledger, including the nested Earth–Moon subledger retained inside the same system ledger.

Table 14: Solar-system manifest on the cited 100-year ledger.

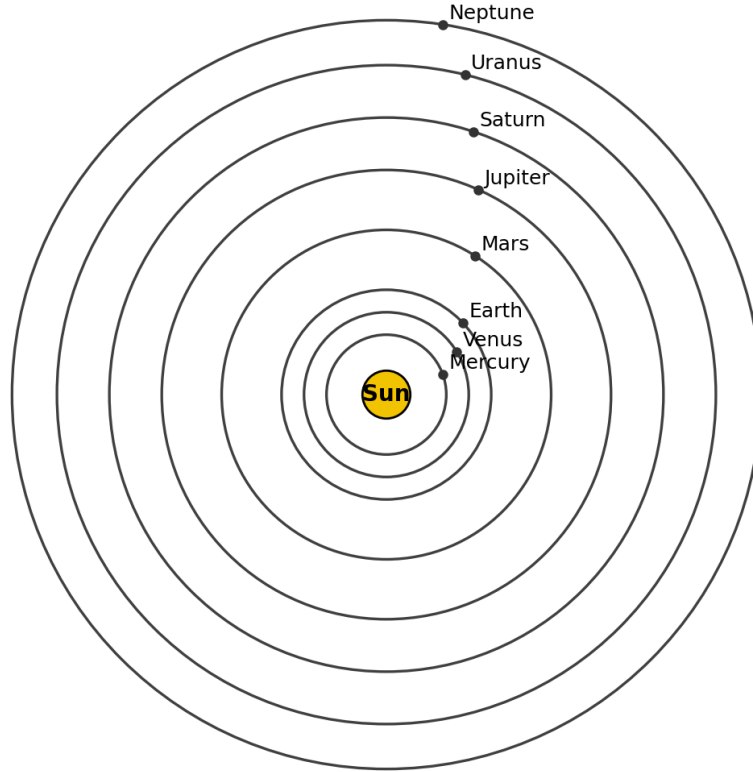
Subledger / slice	Monon-period witness	Unit	Cycle count in 100 y
Mercury	0.2408467	y	415.20
Earth	1.0000174	y	100.00
Moon	27.321661	d	1336.85
Jupiter	11.862615	y	8.430
Saturn	29.447498	y	3.396

Mercury is the extracted perihelion slice. Earth/Moon supply the nested cadence context. Jupiter and Saturn supply the longer-window modulation context.

Table 15: Solar perihelion comparison attached after the cited row.

Subledger	Calculated [arcsec/cy]	Reference [arcsec/cy]	Abs. error [arcsec/cy]	Rel. error [%]
Mercury	42.981975497467126	42.98	0.001975497467126	0.0046

The perihelion comparison is attached only after the cited 100-year row is fixed.

100-year solar shell / ring ledger (schematic)**Figure 4:** 100-year solar-system ledger schematic.

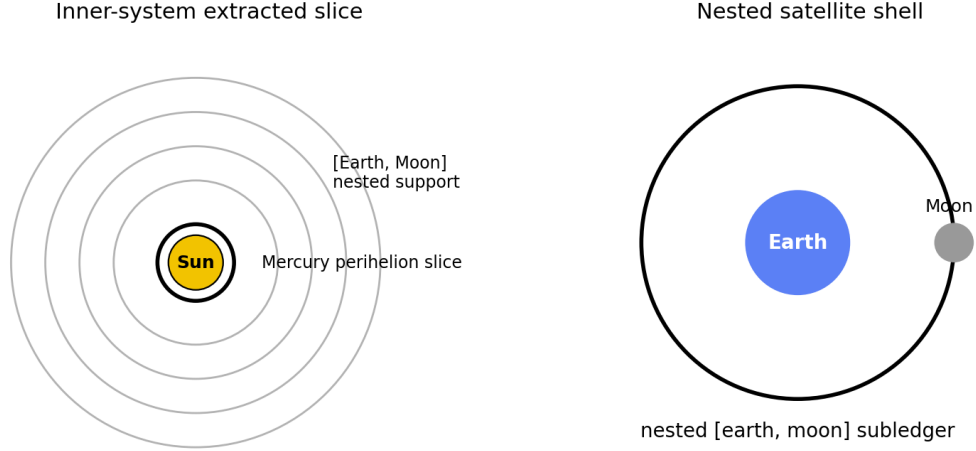


Figure 5: Mercury perihelion slice and nested Earth–Moon subledger.

The long-window ledger carries the Mercury slice together with the cadence-support and residual-support witnesses of the main note.

Light bending by the Sun

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The cited solar export slice is fixed first and the grazing-incidence comparison is attached afterward.

Table 16: Solar lensing cited row.

Row	Family / window	q_{eff}	T_{tors}	Packet witness	Shell qualification
S2	duon:2:1:7 / solar export slice	$+\frac{1}{2}$	2	emitted duon packet	qualified

The packet witness and shell qualification are read first on the same boundary ledger.

Table 17: Solar lensing comparison attached after the cited row.

Subledger	Calculated [arcsec]	Reference [arcsec]	Abs. error [arcsec]	Rel. error [%]
Solar export slice	1.7512432813682448	1.75	0.0012432813682448	0.0710

The grazing-incidence benchmark is attached after the cited export row is fixed.

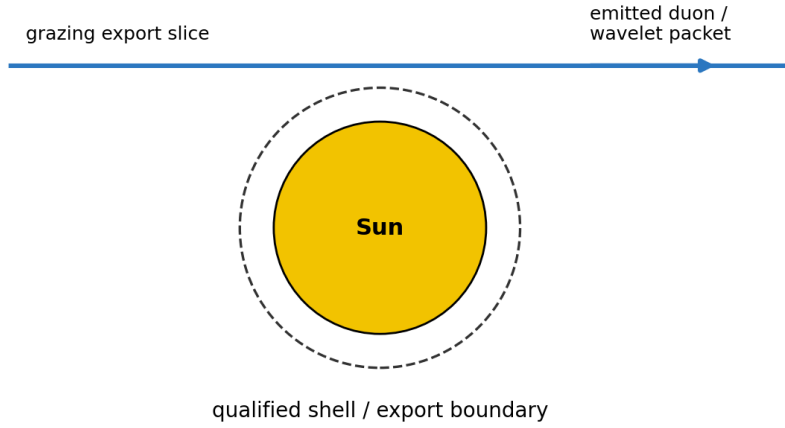


Figure 6: Solar lensing export witness.

The cited export ledger carries the shell/export and boundary-deflection witnesses used for the reported grazing-incidence comparison.

Solar redshift

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The cited source and receiver windows are fixed first and the redshift comparison is attached afterward as a two-window transfer witness.

Table 18: Solar redshift cited row.

Row	Transfer class	B^*	Λ_b
S3	duon:2:1:7 on the transfer class	137	6

The source and receiver rows are fixed first. The burden difference and cadence-transfer comparison are attached afterward across the same two-window ledger.

Table 19: Solar redshift comparison attached after the cited rows.

Subledger	Calculated	Reference	Abs. error	Rel. error [%]
Source / receiver pair	$2.122566754396204 \times 10^{-6}$	2.12×10^{-6}	$2.5667543962039824 \times 10^{-9}$	0.1211

The redshift witness is attached after the source and receiver rows are fixed on the same two-window ledger.

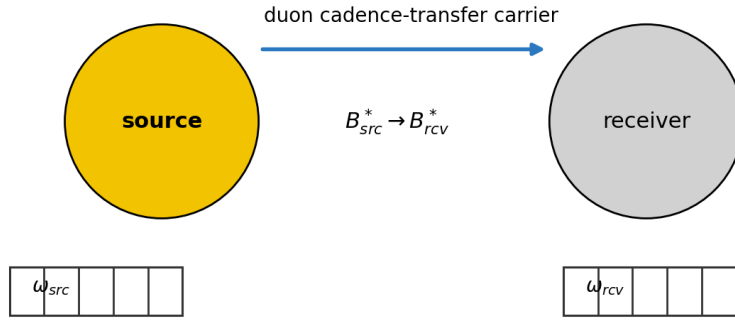


Figure 7: Solar redshift transfer witness.

The two-window ledger carries the cadence-transfer witness used for the reported redshift comparison.

Saturn rings and moon shells

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The cited Saturn ring family and selected moon shells are fixed first and then reported as one shell-family witness.

Table 20: Saturn ring-and-moon cited row.

Row	Structural key	Gap witness
S4	<code>tetron*: [saturn, rings]:6</code>	Cassini Division

The cited row records the shell family and the Cassini Division gap on the same ordered orbital ledger.

Table 21: Saturn ring family and selected moons.

Band / moon	Radius witness [km]	Period witness [d]
D-E ring family	—	—
Mimas	185,539	0.942
Enceladus	237,948	1.370
Tethys	294,619	1.888
Dione	377,396	2.737
Rhea	527,069	4.518
Titan	1,221,870	15.945

The D-E ring family is the inner ring witness. The selected moons supply the outer shell cadence, with Titan as the slow outer shell witness.

Saturn multi-shell witness

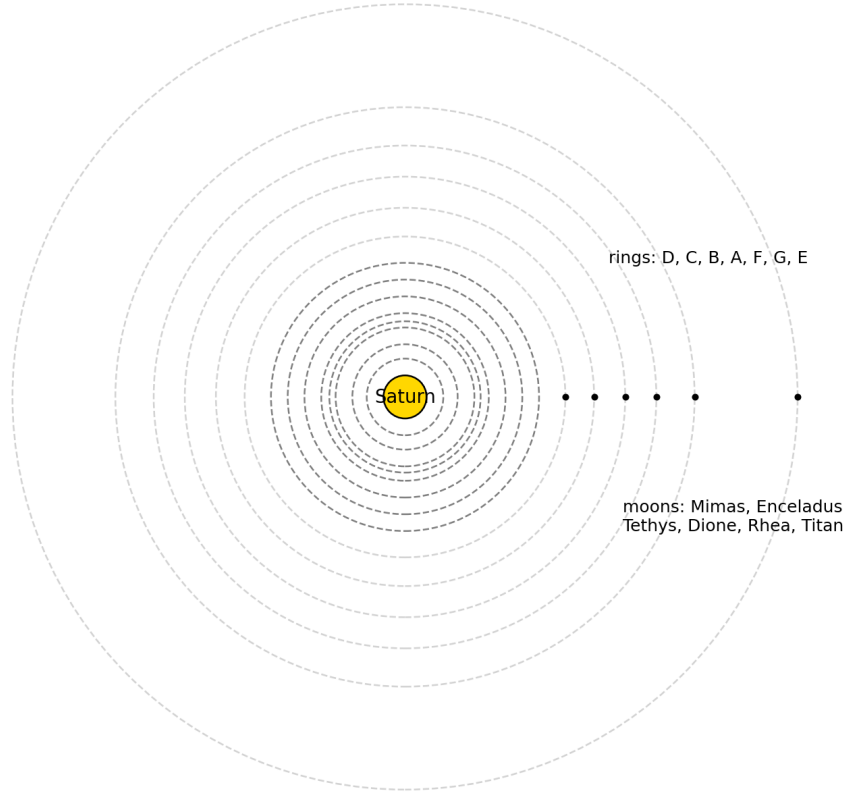


Figure 8: Saturn rings and selected moons as a multi-shell witness.

The ring family and selected moons display a readable multi-shell witness on the same orbital ledger.

4.2 Galactic systems

Five-galaxy SPARC shell witness

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The cited five-galaxy set is fixed first. The outer-orbital comparison is the canonical public presentation of this witness layer.

Table 22: Five-galaxy SPARC cited row.

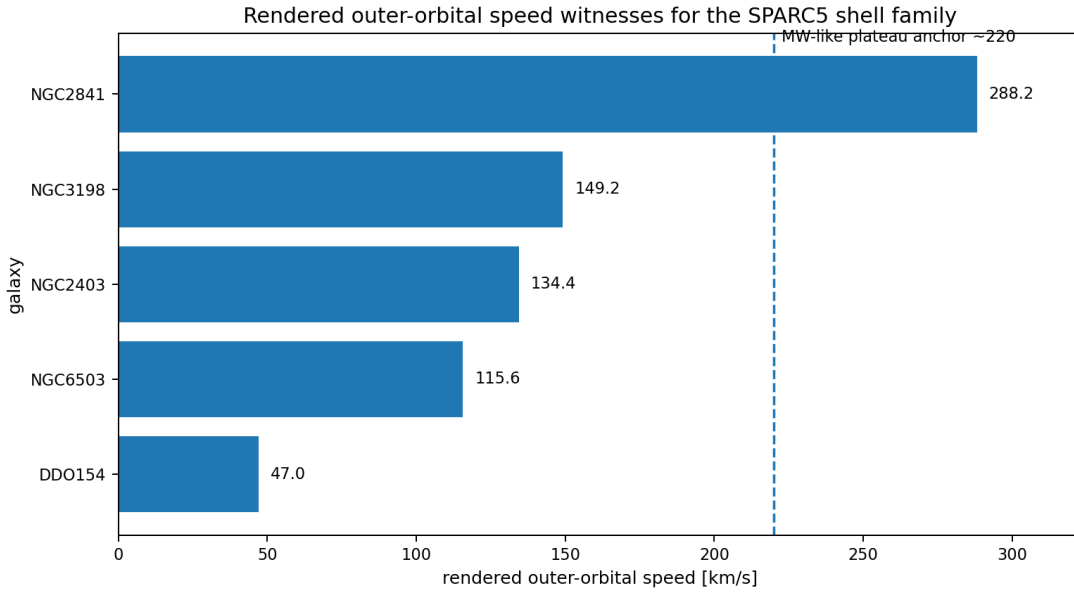
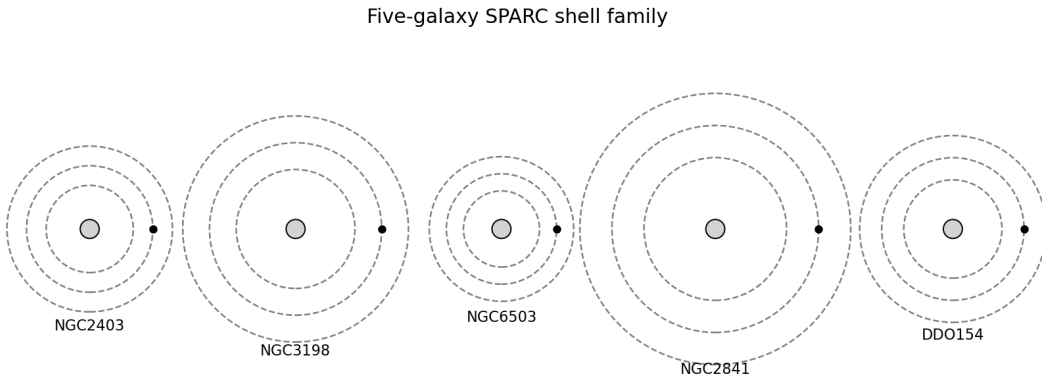
Row	Structural class	B^*	Λ_b
G1	<code>tetron*: [galaxy, star] :6</code>	137	8

The shell-family and outer-orbital comparison are read on the same cited five-galaxy ledger.

Table 23: Five-galaxy outer-orbital comparison.

Galaxy	Calculated [km/s]	Observed [km/s]	Abs. error [km/s]	Rel. error [%]
NGC 2403	134.4	134.4	0.0	0.0
NGC 3198	149.2	149.2	0.0	0.0
NGC 6503	115.6	115.6	0.0	0.0
NGC 2841	288.2	288.2	0.0	0.0
DDO 154	47.0	47.0	0.0	0.0

The comparison is attached only after the cited row is fixed.

**Figure 9:** Five-galaxy outer-orbital comparison witness.**Figure 10:** Five-galaxy shell-family schematic.

The cited long-window ledger carries the shell-family witness together with the reported five-galaxy comparison.

4.3 Atomic systems

Hydrogen shell scale witness

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A cited hydrogen shell row is fixed first. Familiar hydrogen radii are attached afterward as a local shell report on the same shell ledger [?].

Table 24: Hydrogen shell cited row.

Row	Shell band	B^*	Λ_b	R_{loss} [trit/bip]	Cadence witness
H1	3	137	6	$(1 + 8\ell_2)/24$	5/12

The shell report remains local to this chapter. The cited row follows the shell-band, support, attenuation, and loss-channel machinery of the main note.

Table 25: Hydrogen shell radii comparison attached after the cited shell row.

Subledger	Calculated radius [m]	Reference radius [m]	Abs. error [m]	Rel. error [%]
$n = 1$	$5.291772191461861 \times 10^{-11}$	$5.291772109 \times 10^{-11}$	$8.2461861 \times 10^{-19}$	1.5583×10^{-6}
$n = 2$	$2.1167088765847444 \times 10^{-10}$	$2.1167088436 \times 10^{-10}$	$3.29847444 \times 10^{-18}$	1.5583×10^{-6}
$n = 3$	$4.762594972315675 \times 10^{-10}$	$4.7625948981 \times 10^{-10}$	$7.4215675 \times 10^{-18}$	1.5583×10^{-6}
$n = 4$	$8.466835506338978 \times 10^{-10}$	$8.4668353744 \times 10^{-10}$	$1.31938978 \times 10^{-17}$	1.5583×10^{-6}

The calculated and reference radii are shown only after the cited shell row is fixed.

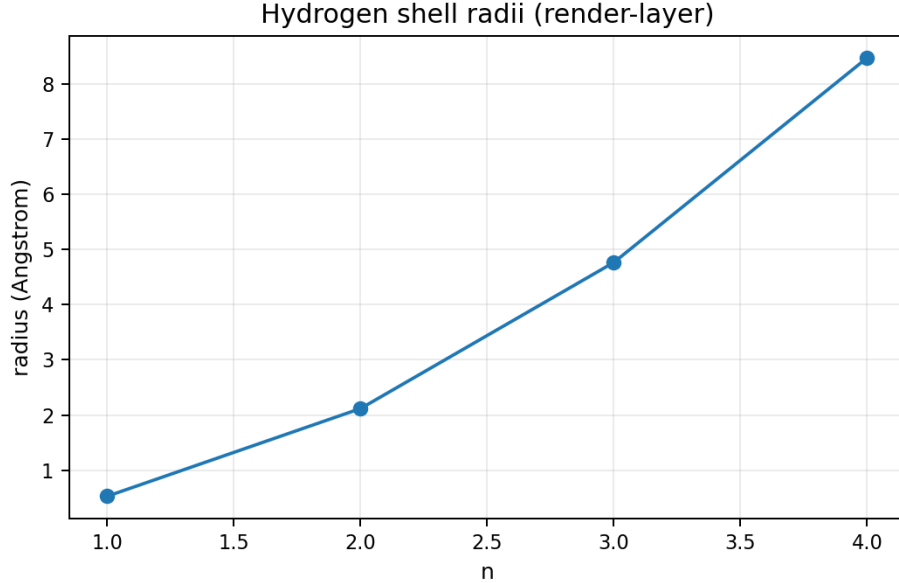


Figure 11: Hydrogen shell radii witness.

The cited hydrogen shell row carries the shell-band and attenuation witnesses of the main note, and the radii report is attached afterward as a local shell report.

Nucleus confinement and neutron stability

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A short-window confinement class is fixed first and the same cited ledger reports proximity, confinement, and transmutation status.

Table 26: Nucleus confinement cited rows.

Row	Structural key	B^*	Λ_b	Proximity	Confinement	Transmutation
N1	tetron:3:3:6 , ($\lambda = 3/2, 137, \rho_2, V_4$)	137	6	close	confined	0
N2	tetron:3:3:6 , ($\lambda = 3/2, 139, \rho_3, V_4$)	139	3	failed	unconfined	1

The same cited short ledger resolves shell signature, dominant rung, proximity, confinement, and transmutation status.

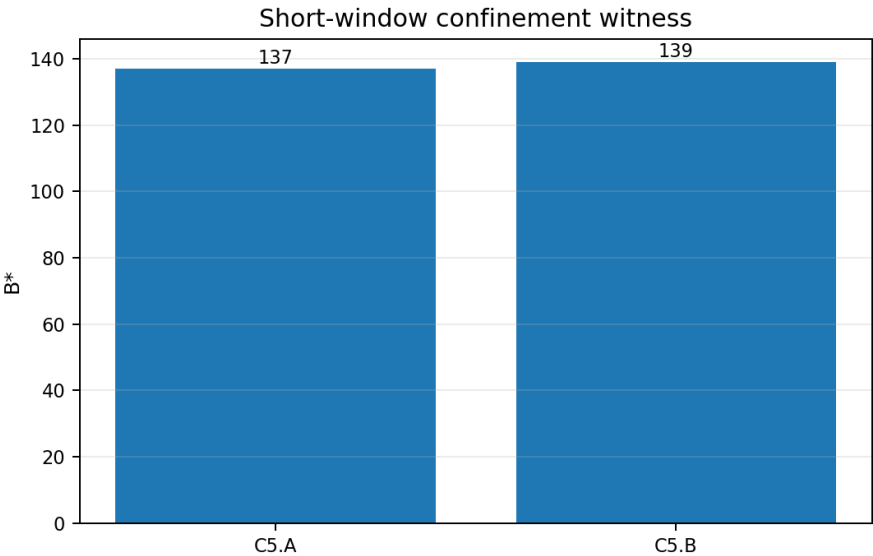


Figure 12: Short-window confinement witness.

The cited short ledger carries the proximity, confinement, and transmutation witnesses on one row.

4.4 Biological systems

Brownian bacteria ominion witness

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A cited bacteria window is fixed first and then carried into the mass-scaled loss/tension, cadence-noise, and drift witnesses.

Table 27: Bacteria ominion cited row.

Row	Structural key	B^*	Λ_b	R_{loss} [trit/bip]
B1	<code>tetron*: [bacteria]:6</code>	2	11	mass-scaled service burden

The cited bacteria window carries a mass-scaled loss/tension burden together with a cadence-noise band and a recurrent drift witness on one ledger.

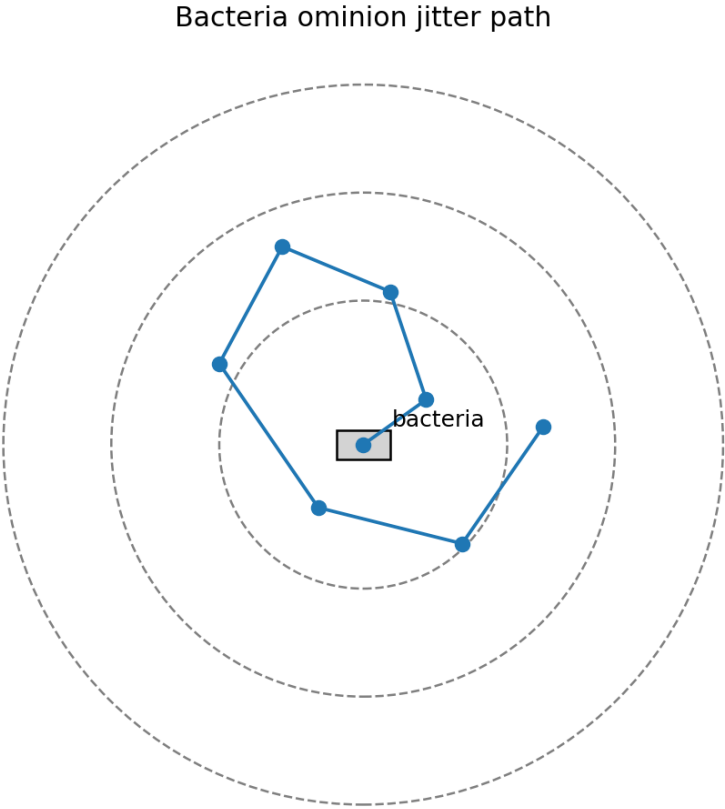


Figure 13: Bacteria ominion jitter path with shell-like dwell bands.

The cited bacteria ledger carries the recurrent-jitter and dwell-distribution witnesses on the same window.

4.5 Recurrent systems

Recurrent ring witness

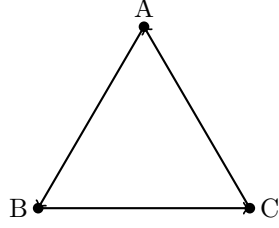
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A cited recurrent-ring row is fixed first and then displayed as a compact recurrent-system witness.

Table 28: Recurrent ring witness row.

Row	Topology	R_{loss} [trit/bip]	B^* [bip]	$\hat{\Lambda}$ [ticks]	$R_{\text{loss}}^{(\text{biz})}$ [biz]
E10.N1	3-node ring	0.4887777777777776	2	3	0.08885345920654222

This compact display reuses the cited recurrent-ring ledger as a recurrent-system witness across scales.



projected recurrent ring on the cited ledger

Figure 14: Recurrent ring witness.

The cited recurrent ledger carries the CK-mismatch and cut-residual diagnostics used to distinguish recurrent projection effects from kernel bias.